Name ____ Date ___________ Period _____

Parent Functions III: 1/x and Asymptotes

TEACHER KEY | Algebra II | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, identify its vertical and horizontal asymptotes, and describe its domain, range, and symmetry.

Vocabulary

Asymptote — a line the graph gets **closer and closer** to but never settles on.

Vertical asymptote — a vertical line the graph **never crosses**. It sits where the function is undefined.

Horizontal asymptote — a horizontal line describing end behavior. It **may** be crossed in other functions.

Reciprocal — the reciprocal of a is **1/a**. Zero has **no** reciprocal.

Part 1 — Build the Table for f(x) = 1/x

Fill in the outputs. One input has no output at all.

x	-2	-1	-1/2	0	1/2	1	2
f(x) = 1/x	-1/2	-1	-2	undef.	2	1	1/2

Why is x = 0 undefined? **you cannot divide by zero**

Part 2 — What Happens Near Zero and Far Out

Follow the pattern in each list, then describe it.

x	1	0.5	0.1	0.01
1/x	1	2	10	100

As x gets close to 0, the outputs grow **without bound**. So there is a vertical asymptote at **x = 0**.

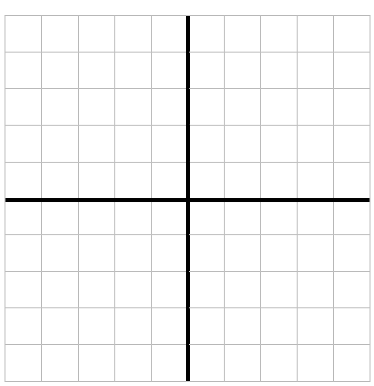
x	1	10	100	1000
1/x	1	0.1	0.01	0.001

As x gets very large, the outputs approach **0**. So there is a horizontal asymptote at **y = 0**.

Part 3 — Graph It

Each square is 1 unit. Plot your table points. You will get two separate branches - do not connect them across the y-axis.

f(x) = 1/x



Record what you see.

Vertical asymptote: **x = 0**

Horizontal asymptote: **y = 0**

How many branches? **2**

Which quadrants? **I and III**

Any intercepts? **none**

Draw both asymptotes as dashed lines.

Part 4 — Key Attributes of f(x) = 1/x

Domain: **all reals except 0** set notation **$\{ x \mid x \in \mathbb{R}, x \neq 0 \}$**

Range: **all reals except 0** interval **$(-\infty, 0) \cup (0, \infty)$**

Symmetry: reflectional across **y = x and y = -x**, and rotational **180°** about the origin.

Maximum or minimum? **neither** Intercepts? **none**

This is the first parent function with **no intercepts** and the first whose domain has a **hole** in it.

Part 5 — True or False

1. A graph may cross a vertical asymptote. **false**
2. A graph may cross a horizontal asymptote. **true**
3. f(x) = 1/x has a y-intercept. **false**
4. The domain of 1/x is every real number. **false**

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: 1/4 = **0.25** 1/(-2) = **-0.5** 1/(1/5) = **5**

2. Solve 1/x = 4. x = **1/4** Solve 1/x = -1. x = **-1**

3. Is there any x with 1/x = 0? Explain.

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4. As x grows very large and negative, what happens to 1/x? **it approaches 0 from below**

5. Fill in the pattern.

x	-0.1	-0.01	0.01	0.1
1/x	-10	-100	100	10

Approaching 0 from the left the outputs go toward **$-\infty$**, from the right toward **$+\infty$**.

6. Why is 1/x the only parent function so far with a gap in its domain?

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Exit Ticket

A function has no intercepts, a vertical asymptote at x = 0, a horizontal asymptote at y = 0, and two branches in Quadrants I and III.

Name it: **f(x) = 1/x** Domain in set notation: **$\{ x \mid x \in \mathbb{R}, x \neq 0 \}$**